

YU YI (NICOLE) KUO

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Education

The Ohio State University

Aug. 2024 - May 2026(Expected)

Master of Science in Computer Science and Engineering

Columbus, OH

Coursework: Computer Architecture, Machine Learning, Parallel Computing, Data Mining, Natural Language Processing

National Ilan University

Sep. 2020 - Jun. 2024

Bachelor of Engineering in Computer Science and Information Engineering

Yilan, Taiwan

Coursework: Data Structure, Data Science, Operation System, Software Engineering, Algorithm, Microprocessor System

Work Experiences

AI Intern

May 2025 - Jun. 2025

NUWA Robotics

Taipei, Taiwan

- Cut memory consumption by up to 80% by implementing and benchmarking hybrid **Retrieval-Augmented Generation** systems using **Milvus** and **BGE-M3** embeddings across multiple **Large Language Model** variants
- Raised natural-language-to-**MongoDB** query conversion accuracy by 13% by engineering text-to-SQL prompt optimizations with **Chain of Thought** prompting and containerized evaluation workflows
- Systematized cross-model performance evaluation by developing automated testing pipelines across **QWen2** and **Llama 3.2**, using semantic similarity and generation-efficiency metrics to identify optimal configurations

Auto Quality Assurance Intern

Jan. 2024 - Jul. 2024

BTSE

Taipei, Taiwan

- Expanded automated test coverage by 56% and reduced execution time by 43% by developing a **Python/Pytest** automation framework for cryptocurrency futures and spot market workflows
- Streamlined feature deployment by 58% by integrating **Postman** API contract checks and **WebSocket** real-time market-data tests into **Jenkins CI/CD** and **GitLab** pipelines
- Shortened defect resolution time by 27% by formalizing **Jira**-based QA triage workflows across a 10+ member team, improving cross-release stability

Software Engineer Intern

Jul. 2023 - Sep. 2023

Neten Digital Consultant

Yilan, Taiwan

- Boosted user engagement by 29% by developing a responsive venue-sharing platform with **Vue.js** and QR-code-based interactions
- Accelerated user authentication by 20% by delivering 5 core features, including multilingual support (2 languages), interactive content display, and validated user authentication
- Enhanced cross-regional user satisfaction by 16% and reduced deployment time by 40% by designing **UI/UX** flows and **MySQL** schemas with a **Go** backend, **Vuetify** frontend, and containerized deployment

Selected Projects

Motion Replacement for Video-Based Remote Physiological Measurement

Apr. 2026

- Reduced heart-rate estimation error by up to **44.92%** in motion-heavy scenarios by building a **2-stage** video preprocessing pipeline for remote physiological measurement
- Improved robustness across **3** supervised backbones by converting 256 x 256 high-motion facial videos into lower-motion samples with frame-by-frame motion replacement and temporal stabilization
- Validated the approach across **2** public datasets and **4** evaluation settings, reaching MAE as low as **3.26** on VIPL-HR and **6.97** on MMPD without modifying downstream model architectures

AI-Driven Recommendation System for FDA Medical Device Safety

Nov. 2025

- Built a recommendation pipeline for **640+** MAUDE publications by using **GPT-4** to enrich MeSH terms from title and abstract text, reducing terminology mismatch in device safety literature discovery
- Sustained recommendation coverage under sparse metadata by enforcing MeSH exact-match retrieval first and backfilling to at least 10 candidates with **MedCPT** cosine-similarity search when exact matches were limited
- Validated and refined ranking quality with 2 domain raters across 60 source papers and top-5 recommendations, and tuned study-design and document-type weighting to surface clinically comparable cases faster

Political Impersonation Detection for Taiwanese Social Media Analysis

Apr. 2025

- Detected political trolling with **75%** accuracy across **1000+** Taiwanese social media posts by building an LLM-assisted impersonation analysis pipeline
- Processed large noisy post collections with a hybrid detection workflow that combined **Gemini API** prompting, weighted feature analysis, and exponential-backoff retry logic
- Raised detection precision from **32%** to **80%** by evaluating zero-shot, contextual, and example-based **few-shot** prompting strategies

Breast Cancer Detection Auxiliary System with Mixed Reality

Mar. 2024

- Built a breast cancer assistive diagnostic system for mammography review and achieved **70%** calcification classification accuracy with an EfficientNetB7-based model
- Benchmarked **6** CNN backbones on CBIS-DDSM crop images and improved training data quality with histogram equalization, OpenCV preprocessing, and flip/rotation augmentation
- Implemented lesion comparison and annotation workflows that surfaced **4** contour metrics on mammograms and delivered a hands-free mixed-reality interface on HoloLens 2

Technical Skills

Languages: Python, JavaScript, Go, SQL, Bash

ML/Data: PyTorch, TensorFlow, OpenCV, RAG, Prompt Engineering, Milvus (Vector DB), Faiss, MongoDB, MySQL

Tools: Git/GitLab, Jenkins, Docker, Postman, Jira, CI/CD, REST APIs, Websocket, Pytest